

Machine Learning in the Retail Sector

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Introduction

Machine Learning (ML) has emerged as a transformative technology retail sector with no exception. With its ability to analyze vast amounts of data, uncover patterns, and make predictions, ML has revolutionized various aspects of retail operations, from demand forecasting and inventory management to personalized marketing and customer service.

In this paper, we delve into the application of ML techniques in the retail sector, aiming to provide a comprehensive understanding of its potential benefits and challenges. Through an extensive review of existing literature and case studies, we explore how ML can be effectively utilized to enhance decision-making processes, optimize resource allocation, and drive business performance in retail settings. Furthermore, we critically examine the limitations and constraints that may hinder the widespread adoption of ML in retail operations, addressing concerns related to data quality, model interpretability, and organizational readiness.

1. Machine Learning for Demand Forecasting

Most of the retailers use demand forecasting with fixed, rule-based systems to predict what they need in which seasons and manage orders and supply accordingly. This approach works well with small scaled businesses and where the amount of product can be predicted. But at the large scale managing such orders would be very challenging. By predicting the demand of sales, the retailers can minimize the stocks that could be out of stocks and prevent the excessive stocks. This not only improves profitability by reducing lost sales and holding costs but also enhances customer satisfaction by ensuring products are available when needed.

Data Requirement

In the retail sector, the number of demand for sales depends on many factors such as seasonality, time, historical data, promotional activities, market trends, Inventory levels and weather data etc.

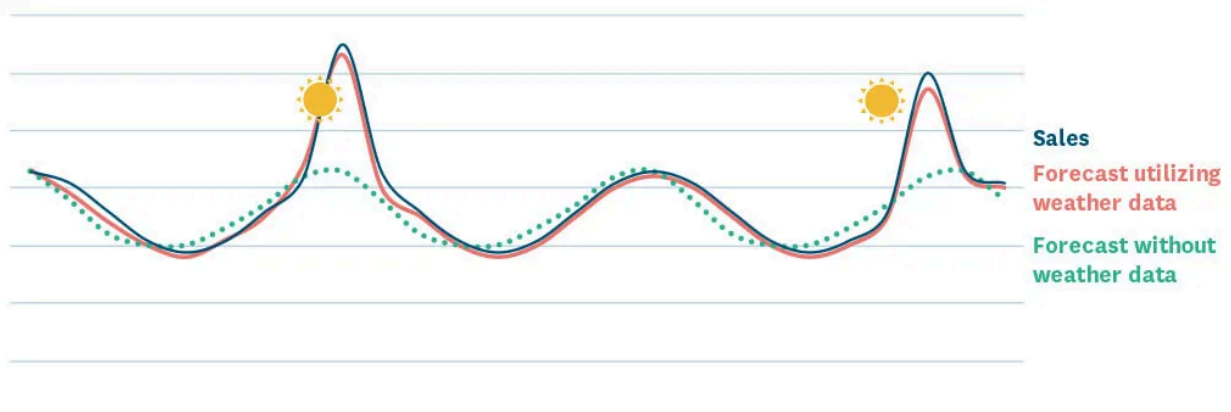
Reason for Implementing ML

The most common method for sales prediction is usually done by time series analysis.

Since in this time analysis we can only consider the attributes like weekly, monthly time sales data. By this analysis we can capture the seasonality also. Only the time is not an attribute that is going to affect the sales. This time analysis only predicts the demand on the basis of time. There are other factors like promotional activities, market trends, weather, local events, and other external factors that will impact the future sales demands. These factors can be combined with time to predict more accurately the number of sales using machine learning models.



1. Demand for this product increases when its price drops, but the increase is bigger when the product's price drops to be the lowest in its category.



2. Impact of weather and other external factors on sales in their retail demand forecasting.

Machine Learning Algorithms:

1. Random Forest

Random Forest is an ensemble learning technique that combines multiple decision trees to make predictions. This model was used because it handles non-linear relationships, making it robust for demand forecasting.

2. Linear regression models establish a linear relationship between input features and the target variable (sales volume).

3. Artificial Neural Network (ANN): Can capture complex relationships between features.

Case Study: Demand Forecasting at Walmart

Walmart, a retail giant, leverages ML algorithms to forecast demand across its vast network of stores. By integrating diverse data sources such as weather data, social events, and historical sales data, Walmart optimizes inventory levels, minimizes stockouts, and enhances overall supply chain efficiency.

Limitations

While ML-based demand forecasting offers significant advantages, it may encounter challenges during extreme events such as pandemics or natural disasters, which disrupt historical patterns. Moreover, substantial data and computational resources are required for effective implementation, posing a potential barrier for smaller retailers.

2. Customer Segmentation

Customers are grouped into clusters so that they can be identified as loyal customers, random infrequent shoppers, customers with potential to become loyal customers, big spenders, small spenders etc. There is no labeled data in this problem but based on the spending of customers, how often they shop and when is the last time that they have shopped some conclusions are drawn to decide the marketing and sales strategies. For example, ease of communication is important for loyal customers and it is assumed that they have the need for recognition for their loyalty. Another strategy is to decide the shipping policy based on customer clusters. For example, if the customer is from the top tier of loyal customer clusters, they get free shipment. If they are from low or mid-tier, they will pay or they will benefit from free shipment depending on the shopping case. If a customer is identified as potentially loyal, they will receive retaining benefits such as special discounts based on volumes or bundled products designed for increasing sales purposes. Depending on the label, if the customer is a rare and small spender, it is advised not to invest time, money and energy for these customers from a business perspective. These cluster labels could be used in communication, promotions, shipment policy, special vouchers, and discounts as marketing strategies.

Machine Learning

Data Collection: Each data point corresponds to one transaction with the details of transaction ID, date, product with its stock codes, quantity, price and customer identifier and country of residence of customers. Transaction ID is the transaction number, uniquely assigned to each transaction.

StockCode is the product code, uniquely assigned to each distinct product.

Description is the name of the product.

Quantity is the quantity of each product purchased per transaction.

TransactionDate is date and time together, the day and time showing when each transaction happens.

UnitPrice is the product price per unit.

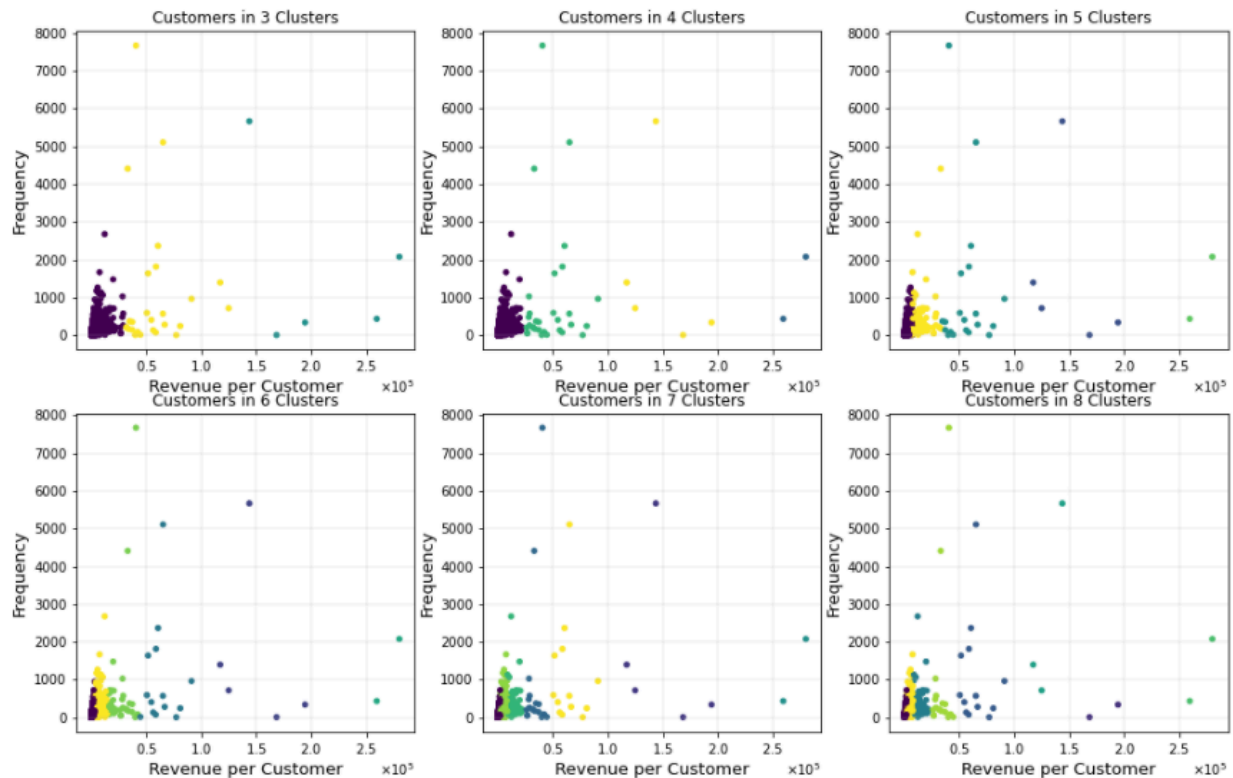
CustomerID, a uniquely assigned number to each customer.

Country is the name of the country where each customer resides. Labels are cluster numbers per customer.

ML Model:

K-means is an unsupervised learning algorithm, meaning it does not require labeled data for training. In this case, where we don't have predefined customer segments.

The clusters generated by k-means are easily interpretable, as each cluster is represented by its centroid, which can be understood as the "average" member of that cluster. We can infer this from the below given fig. For that we have to experiment with finding the number of clusters. Once we have clusters with suitable boundaries, then customers can be labeled with different loyal batches.



Cluster visualization

Source: <https://www.europeanpublisher.com/en/article/10.15405/ejsbs.316/images/12>

Case Study: Amazon

Amazon uses ML to segment its customer base. By understanding shopping patterns, they personalize recommendations, promotions, and product offerings. Segments include high-value customers, dormant users, and churn-risk individuals. [\[source\]](#)

3. Increasing Revenue from Low-Spending Accounts

A big retailer company has mainly three types of spending accounts or customers. One is a high-spend account that is assumed to be a big company, other is medium-spent accounts and low-spending accounts. Low spending accounts are those accounts that have low monetary purchase or lower frequency of purchase of items. These accounts have a very low contribution margin for retailers revenue. Identifying strategies to engage these customers effectively can lead to increased sales and improved customer loyalty. In this section, we propose a machine learning (ML) approach to address this challenge.

Data Collection: historical transaction data for all customers, including purchase dates, order values, and associated companies.

Recency: How recently the customer made a purchase.

Frequency: The number of times the customer has purchased.

Monetary: The total value of purchases made by the customer.

Start Date: When the customer started purchasing.

Density: The average time between purchases.

Year-over-Year Revenue: Compare revenue generated by the customer in the current quarter with the same quarter in the previous year.

Machine Learning Models : these three supervised ML models can be applied :

Logistic Regression: Logistic Regression is suitable for binary classification problems, where the prediction can be divided into two classes. In the case where I want to predict whether a low-spending account will make a purchase or not. This model can be useful. Even this model can interpret results by examining features.

Artificial Neural Network (ANN): Can capture complex relationships between features.

Random Forest: Ensemble method that combines multiple decision trees.

Train these models on historical data, using the features as input and the target variable (purchase/non-purchase) as the output.

Reinforcement Learning: Reinforcement learning is a type of ML which rewards actions by learning from its past successes and mistakes. Through trial and error, reinforcement learning finds the best path to a reward. For Starbucks, this reward is a purchase.

Real-World Example: Starbucks

Starbucks uses ML to analyze transaction data and identify low-spending customers. By offering personalized promotions and loyalty rewards, they successfully increase revenue from these segments. [\[source\]](#)

Limitations

ML cannot replace human intuition entirely. Understanding context and building relationships with low-spending customers require a human touch. Additionally, privacy concerns may limit data availability.

Where Machine learning can not be applied

1. Offline retail often lacks digital data compared to online platforms. ML models depend on large, high-quality datasets, but collecting and maintaining such data in physical stores can be challenging. In-store transactions, inventory levels, and customer behavior are not always seamlessly captured and digitized.
2. ML algorithms often require real-time data for accurate predictions. In offline retail, real-time data updates are harder to achieve due to manual processes and delays.
3. Physical Constraints: ML excels in digital environments but faces challenges in physical spaces (e.g., store layouts, inventory management). Integrating ML seamlessly into brick-and-mortar stores is not straightforward.

Conclusion

Machine Learning is a powerful tool for retailers, but its success depends on thoughtful implementation, data quality, and understanding its limitations. By considering both the benefits and challenges, retailers can harness ML to create value for their end-users while driving business growth.

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